

Week 6 Project Guide

Probabilistic Models & Bayesian Inference

Role Requirements · Notebook Walkthrough · Submission Guide

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How to use this document: Read the **Your Role** section first — it sets the business context for every decision you will make in the notebook. Then work through the **Notebook Walkthrough** section by section. Before you submit, run through the **Completion Checklist** and the **Submission Requirements** to make sure nothing is missing. All links in this document are clickable.

1 YOUR ROLE

The business scenario you are operating in for this entire assignment

You are the **Data Scientist who deployed last week's XGBoost churn model**. Three weeks in, the VP of Retention flags two concerns: (1) “When you say a customer has a 73% churn probability — what is the margin of error on that number?” and (2) “We just launched a new contract tier with only 40 customers enrolled. Your model has never seen this segment — how far should I trust its output?”

Responsibility	What it means in practice
Replace point estimates	Parts 1–2: compute MLE, MAP, and full posteriors for churn rate by contract type. Report 94% HDI — not a single number.
Quantify sequential evidence	Part 2: implement <code>update_posterior()</code> and show the VP exactly how many customers are needed before the inference is reliable.
Model feature dependencies	Parts 3–4: fit a Multivariate Gaussian, a Bayesian Network, and a Markov Random Field (MRF) to the same Telco features, and explain what each tells you that a single coefficient cannot.
Forecast with honest uncertainty	Part 5: fit a GP to Mauna Loa CO ₂ and define the model confidence boundary — the horizon at which the 95% credible band exceeds 5 ppm.
Validate a Bayesian posterior	Part 6: build a Bayesian Logistic Regression using PyMC/NUTS, diagnose convergence via R-hat and ESS, compare the posterior to the frequentist MLE, and save the fitted trace as a .pkl artefact.

Key challenge: These are not just modelling exercises. Each part answers one of the VP's two questions directly: (1) uncertainty quantification on point estimates, and (2) reliability of inference on small segments. A model that gives you a number without a distribution is giving you an incomplete answer — this week you replace every such number with a posterior.

2 DATASETS

Two datasets, six parts — no new EDA required for Telco

Dataset	Source	Used in	Notes
Telco Customer Churn	Kaggle	Parts 1, 2, 3, 4, 6	7,043 rows · 21 cols · same CSV from Week 5. No new EDA — focus is on probabilistic methods.
Mauna Loa CO₂ (monthly mean)	statsmodels built-in	Part 5 only	521 months (1958–2001) · zero download · 3 lines to load. No CSV, no Kaggle.

Mauna Loa data loading (3 lines — no CSV):

```
from statsmodels.datasets import co2

df = co2.load_pandas().data.resample('ME').mean().dropna()

t = (df.index - df.index[0]).days.values.reshape(-1, 1) / 365.25

y = df['co2'].values
```

Design rationale: Telco carries the business narrative across five of six parts — students already know the data, so cognitive load stays on the probabilistic concepts. Mauna Loa is used only for the GP section because it has the exact structure GPs are designed for: a continuous temporal signal with trend, seasonality, and measurement noise that *requires* composite kernel engineering and produces visually dramatic uncertainty bands on extrapolation.

3 HOW TO USE THE NOTEBOOK

Step-by-step guide for opening, running, and completing the assignment

- Open the notebook shared with you in Google Classroom and load it in Google Colab.
- Run the setup cell first to install all required packages (pymc, arviz, pgmpy, scikit-learn, statsmodels, scipy). This takes ~3 minutes on a fresh runtime.
- Work through Parts 1–6 in order — each part's outputs feed into later parts.
- When done, submit the completed notebook (.ipynb), the saved PyMC trace (telco_bayes_lr_v1.pkl), and your one-page reflection back through Google Classroom.

4 SUBMISSION REQUIREMENTS

Exactly what to submit, how, and what will be checked

4.1 Files to Submit

File	Description	Required?
W6_Probabilistic_Models_Assignment.ipynb	Your completed assignment notebook. Must be fully executed — all cells run top-to-bottom with all outputs, posterior plots, credible-band plots, and trace plots visible. All SELF-CHECK cells pass. All 🐞 Reflect cells filled in.	✓ Required
telco_bayes_lr_v1.pkl	The fitted PyMC InferenceData trace saved using pickle.dump(). Must load without errors using pickle.load() and contain all sampled chains. This mirrors Week 5's joblib serialisation habit — probabilistic models have artefacts too.	✓ Required
Reflection PDF or .md (≤ 1 page)	Answer in your own words: 'Give one concrete example from this assignment where the fully Bayesian answer would have changed a decision you might have made using only the MLE or a p-value. Explain the mechanism — was it prior information, uncertainty quantification, sequential updating, or the Bayesian-vs-frequentist comparison?'	✓ Required

4.2 Notebook Quality Standards

- All cells are executed in order from top to bottom — no skipped cells, no cells with errors in the output.

- Every `# YOUR CODE HERE` has been replaced with working code.
- Every 🐞 **Reflect** markdown cell contains your own reasoning — not a copy-paste from the session notebook or an AI-generated answer.
- All **SELF-CHECK** assert cells pass without errors.
- Trace plots (`az.plot_trace`) are visible for Part 6 — not empty axes.
- The Part 6 prior-sensitivity overlay (Q20) and the Bayesian 94% HDI vs. frequentist (sklearn) MLE comparison (Q21) are visible, with specific numeric values stated.
- Posterior HDI intervals are reported with specific numeric values — no placeholder text.
- Code is clean and readable — no commented-out dead code blocks left behind.

Do not submit: The raw CSV data file (too large), the session solution notebook, any `.pyc` files or `__pycache__` directories, or files generated by Jupyter checkpoints (`.ipynb_checkpoints/`). The `.pkl` trace file is the only binary artefact — do not submit joblib files from Week 5 again.

5 COMMON MISTAKES TO AVOID

Patterns that cause lost marks or incorrect results

✗ Not scaling features before NUTS sampling (Part 6)

PyMC's NUTS sampler assumes the posterior geometry is roughly spherical. Unscaled continuous features (tenure in months, MonthlyCharges in dollars) create elongated, highly anisotropic geometries that the sampler traverses very slowly — resulting in high autocorrelation, low ESS, and divergent transitions. Always scale to zero mean / unit variance using training-set statistics only before passing features to the model.

✗ Choosing uninformative flat priors for logistic regression (Part 6)

`pm.Flat()` on logistic regression coefficients creates an improper posterior — NUTS will diverge or produce nonsensical samples. Use weakly informative priors: `Normal(0, 2)` for β coefficients and `Normal(0, 5)` for the intercept. These encode 'most coefficients are between -4 and $+4$ on the log-odds scale' while remaining dominated by the data at $n \approx 5,000$.

✗ Ignoring $R\text{-hat} > 1.01$ or bulk-ESS < 400 (Part 6)

Reporting results from a non-converged chain is equivalent to reporting a model trained on corrupted data. If any parameter has $R\text{-hat} > 1.01$, do not proceed to posterior analysis. Investigate the trace plot, check for divergences, consider increasing `target_accept` or re-scaling features. $\text{ESS} < 400$ means your HDI estimate has high uncertainty itself.

X Reporting MLE or MAP as 'the Bayesian answer' (Parts 1–2)

MLE and MAP are both point estimates. The fully Bayesian answer is the entire posterior distribution — plot it, report the HDI, and use Monte Carlo to answer $P(\theta_A > \theta_B)$. Reflecting only on the MAP estimate for Q1 will lose marks: the point of the exercise is the overlap (or non-overlap) of the full posteriors.

X Using the wrong marginalisation formula (Part 3)

To marginalise out TotalCharges from the 3D covariance matrix, you simply extract the 2×2 upper-left submatrix of $\Sigma_{\{3D\}}$. Do not re-fit a new 2D Gaussian — analytically marginalising means taking the corresponding block of the covariance. Verify that the result matches the directly fitted 2D Σ as a numerical check.

X Writing generic Reflect answers

Answers like 'the posterior gave more information' or 'the GP had uncertainty bands' earn no marks. Reflect answers must be specific: quote the numeric HDI, name the feature, cite the R -hat value, identify the exact n at which the decision threshold was crossed. E.g. 'At $n=87$ the posterior $P(\theta > 0.25)$ first exceeded 0.90 — 43 observations earlier than the frequentist z-test sample size of 130 would have required.'

Remember: The combination of correct code + written reasoning is what separates a model that informs decisions from one that sits in a notebook. A perfect MCMC trace with empty Reflect cells will not pass. A thoughtful reflection connecting the HDI to the VP's actual question — even with a minor code bug — is much better.